Development of Online Students' Marksheet Generation and Disrtibution Management System

A Mini-Project submitted for the partial fulfillment of Advanced Database Management System Laboratory (CS15204)

By

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Ravangla – 737139, Sikkim
December 2025

CERTIFICATE

This is to certify that the project report entitled "**Development of Online Students' Marksheet Generation and Distribution Management System**", submitted by **Yash Gupta** is an authentic report for the work carried out under my supervision as a part of Advanced Database Management System Laboratory during the Odd Semester, 2025-26 at National Institute of Technology Sikkim, Ravangla–737139, Sikkim.

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ABSTRACT

The “Online Students’ Marksheet Generation and Distribution Management System” is a website designed to streamline the process of allowing students to view their marks attained in Institute’s examinations. The project aims to replace the traditional manual methods of accessing academic records which is often very limited both in time and accessibiltiy also requiring atleast one staff member busy to allow students to verify their marks and grades obtained so far. The online system (as reffered to MGDM, short for Marksheet Generation and Distribution Management) provides a centralized and secure platform for three key users: Admins, Teachers and most importantly Students.

Admins have overarching control with capabilitites for bulk registration of students and adding subjects using standard spreadsheets thereby reducing their load of manually typing everything onto website. Teachers are also provided with an efficient interface to upload student marks for various assesments using standard spreadsheet format heavily utilised for storing and keeping track of student marks, thereby relieving pressure on Teachers to learn and use new software and/or logging every time they need to update any student’s marks. Students are allowed 24 x 7 access to view their obtanied grades in all subjects semster-wise and also check their SGPA whenever they want alongwith the facility for downloading their results in a pdf format.

The application is built on the robust Django framework with MariaDB for the database, ensuring scalability, security and maintainibility. The successful implementation of this project provides transparent, accurate and efficient solution for managing marksheet generation and distribution to students.

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1. INTRODUCTION

Today, with the increase in number of students and as Institutes gets larger, the timely dissemination of exam results to each and every student continues to become a hassle and a bottleneck to student satisfaction when it comes to accessibility. Results are generally only availble to be verified for a limited amount of time on a piece of paper attached to a noticeboard with limited information. Further, the results get disclosed to everyone which could potentially beome a stress factor to under-performing students. Further, collecting marksheets from the examination cell is also not efficient, requiring both student and staff member to be present in the office at the same time and wait for staff to search through student's marksheet.

The "Online Students' Marksheet Generation and Distribution Management System" is a response to counter these challenges and cater to needs of student thereby increasing student satisfaction and increasing overall efficiency of the entire system. By providing a centralized platform, the system ensures a single secure source for students to

2. PROBLEM DESCRIPTION

The existing manual system marksheet generation suffers from several critical drawbacks:

- Inefficiency and Time Delays: The process of collecting marks from multiple teachers teachers for numerous subjects is slow and cumbersome, leading to significant delays in the final publication of results.
- High Risk of Human Error: Manual data entry and calculation of grades, SGPA, and CGPA are highly susceptible to human error, which can have serious consequences for students.
- Lack of Centralization: Academic data is often scattered across various different departments in spreadsheets and physical documents, making it difficult to manage, secure and retrieve historical records.
- Limited Accessibility for Students: Students typically have to physically visit the instituition to view their results or collect their marksheets which can be incovinient and inefficient.

3. TOOLS AND TECHNOLOGIES USED

- **Programming Language:** Python
- **Backend Framework:** Django (Python)
- **Database:** MariaDB
- **Frontend Technologies:** HTML, CSS, JavaScript
-
- **Key Python Libraries:**
  - **pandas:** For reading and processing of spreadsheet (.xlsx, .odf) files.
  - **Reportlab:** For dynamic generation of professional, high quality PDF documents
  - **qrcode:** For creating qr codes to be displayed in the downloadable PDFs.
  - **Mysqlclient:** Database connector between Django and MariaDB.

4. PROBLEM FORMULATION

4.1 OBJECTIVE:

The primary objectives of this project are:

- To design and develop a secure, multi-user web application for managing student academic records.
- To provide distinct, role-based interfaces for Administrators, Teachers, and Students.
- To automate the process of marks calculation, grade assignment, and SGPA/CGPA computation.
- To enable bulk data entry for student and subject registration via spreadsheet uploads.
- To implement a secure administrative workflow for the verification and publication of marksheets.
- To generate secure, on-demand, and downloadable PDF marksheets with features like watermarks and QR codes for authenticity.

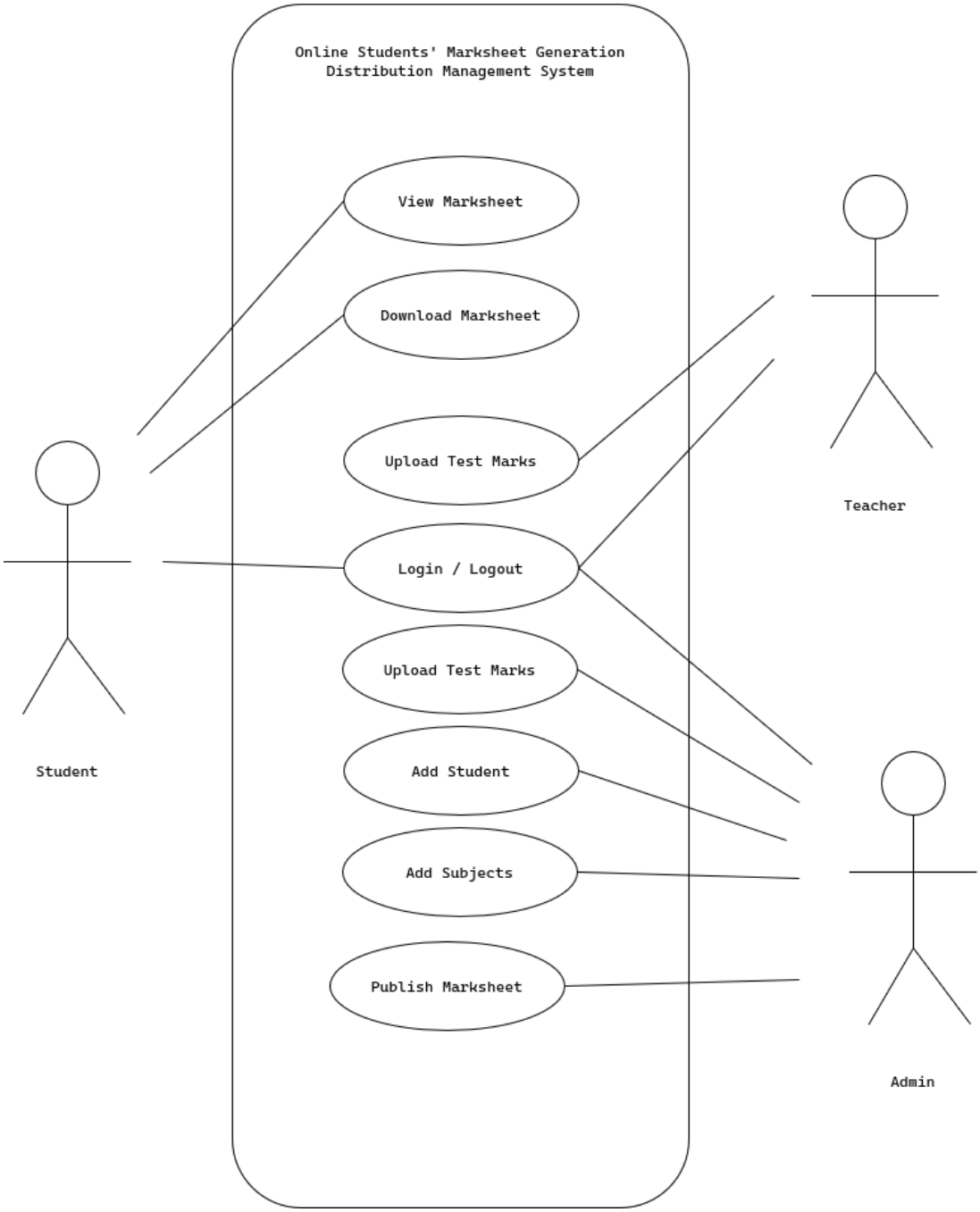
4.2 PROPOSED SYSTEM

The proposed system is a centralized web application built on the Django Model-View-Template (MVT) architecture. It features three distinct user roles:

1. **Administrator:** The superuser with complete control over the system. They manage courses, departments, and user accounts. They are responsible for the bulk upload of student and subject data and hold the final authority to publish marksheets for student viewing.
2. **Teacher:** Faculty members can log in to a dedicated portal where they can view the tests they are assigned to, upload marks for their students using a standardized Excel/ODS template, and view or edit previously submitted marks.
3. **Student:** Students can log in securely to view their results. After login, they are prompted to select a semester, after which a detailed provisional grade card is displayed. They have the option to download this grade card as a professionally formatted and secured PDF.

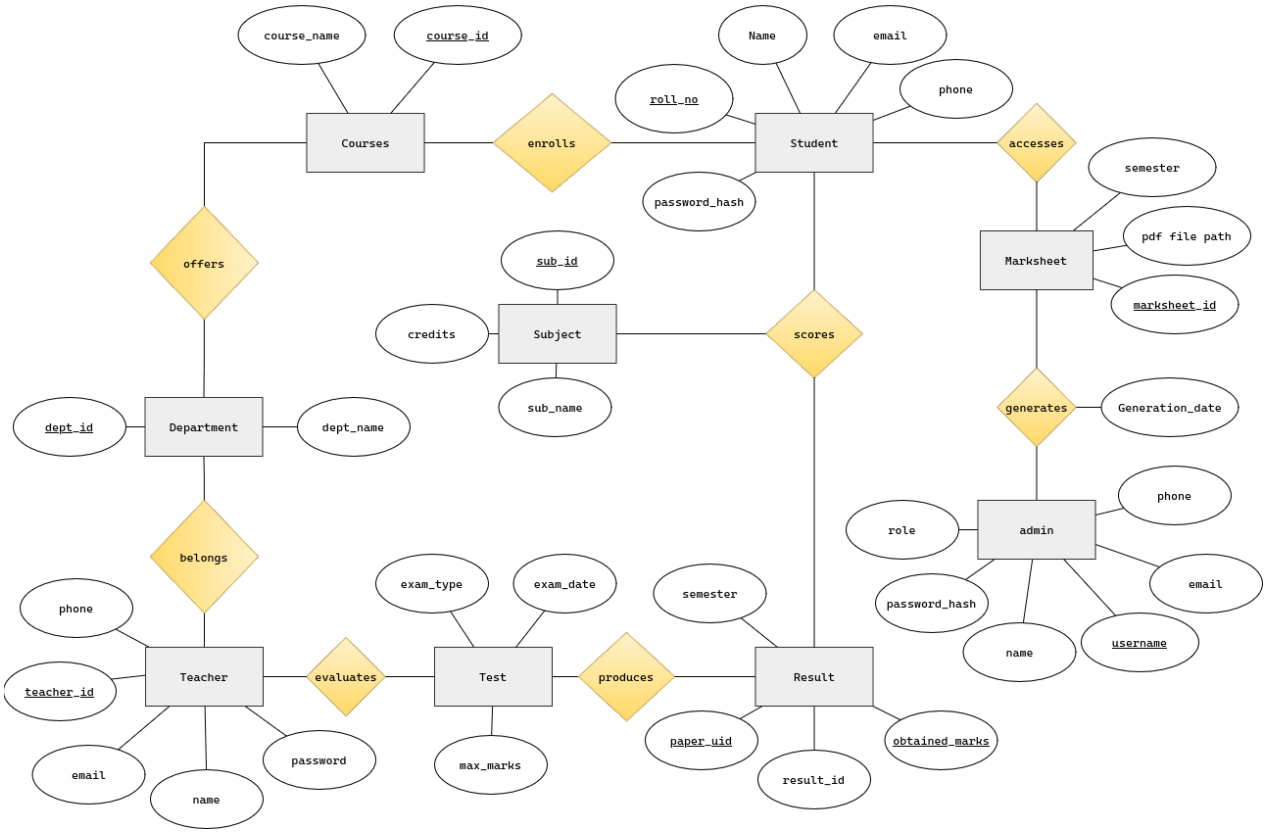
4.3 USE CASE DIAGRAM

Use Case Diagram

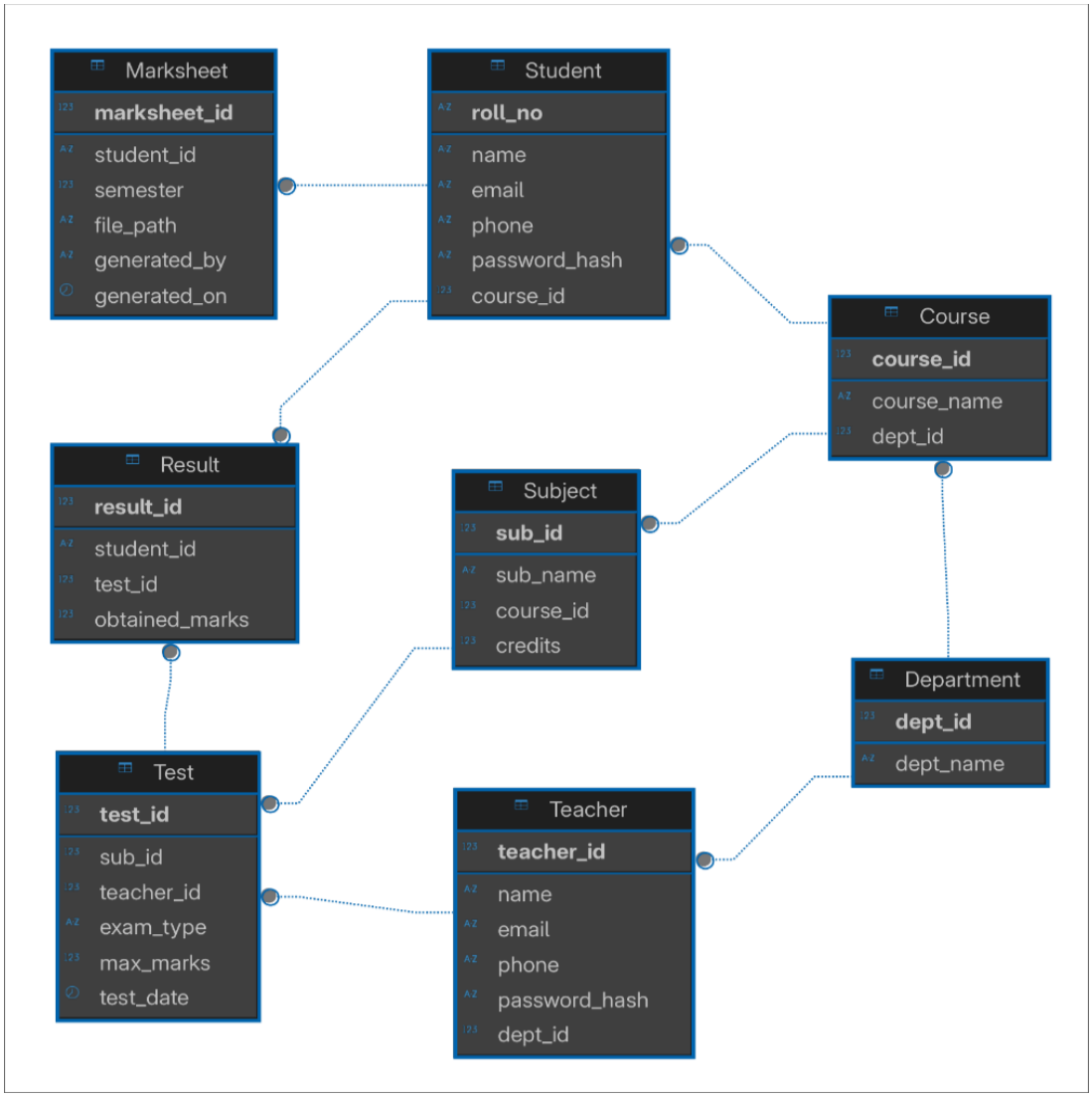


4.4 ENTITY RELATIONSHIP DIAGRAM

ER Diagram



4.5 DATABASE SCHEMA



4.6 IMPLEMENTATION

The system is implemented using the Django MVT (Model-View-Template) architectural pattern:

- Model (models.py): This layer defines the database schema as python classes. Each class maps to a database table, and each attribute maps to a column. It serves as the single, definitive source of the data structure.
- Views (views.py): This is the logical core of the application. Views are Python functions that handle incoming web requests. They interact with the models to fetch or save data from the database, perform calculations (e.g., SGPA/CGPA), and then pass the processed data to the appropriate template for rendering.
- Templates (*.html): This layer handles the presentation and user interface. Templates are HTML files with a special Django syntax that allows for the dynamic insertion of data passed from the views, as well as control structures like loops and conditionals.

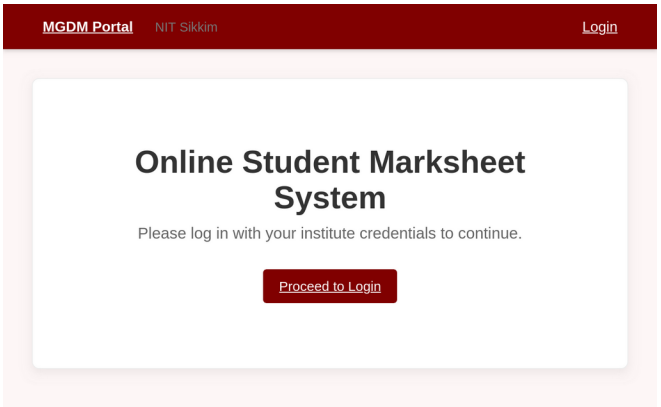
4.7 DESCRIPTION OF MODULES

The application is divided into several key functional modules:

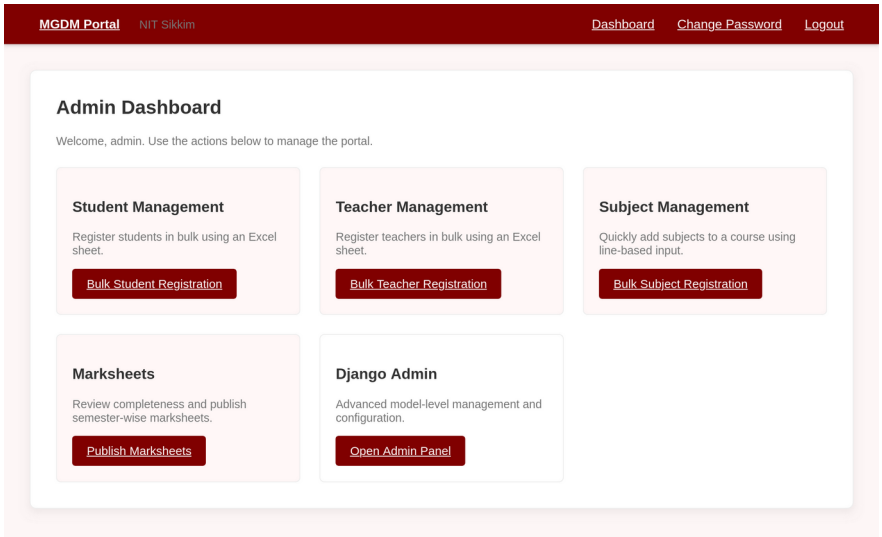
1. Authentication Module: Handles user login, logout and password management. It uses Django's built-in authentication system to securely manage user sessions and permissions, ensuring that users can only access pages and data relavant to their role.
2. Admin Module: A suite of tools to be used by the administrator. This includes the bulk upload interfaces for uploading the student and subject data directly using spreadsheets. And allowing admin to review and publish marksheets as required which then allows students to see their marksheets.
3. Teacher Module: A portal for faculty, it provides a dashboard listing all tests for assigned subjects. For each individual test a teacher can upload marks from a spreadsheet, view the existing marks for each student and make individual edits if required.
4. Student Module: This part of the application allows students to view and download their marksheet. Students have to first login via assigned username and password after which they are allowed to see a list of semester for which marksheets are available and download them.
5. Marksheet Generation Module: This is the backend module that contains the core logic of converting all marks scored by students and converting them to grades, calculating Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA). This module uses ReportLab library to create secure, professionally formatted PDF documents with dynamic data, watermark and QR codes which allow student to scan and verify their obtained grades.

5. EXECUTION SNAPSHOTS

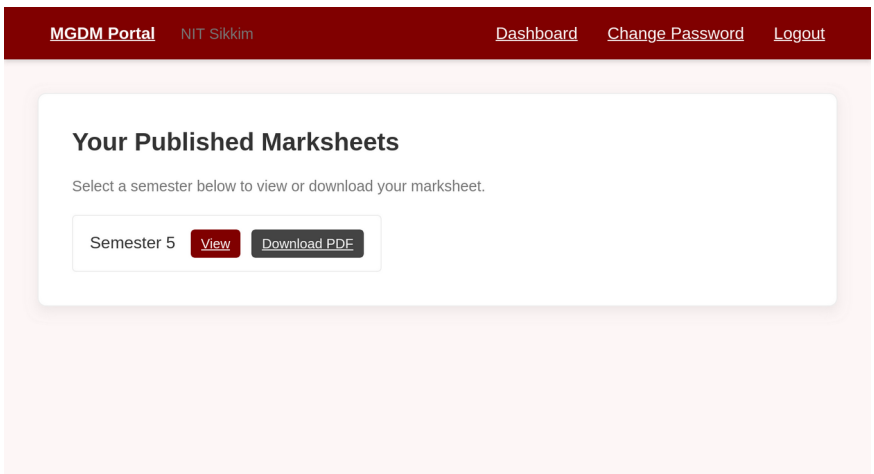
Home Page:



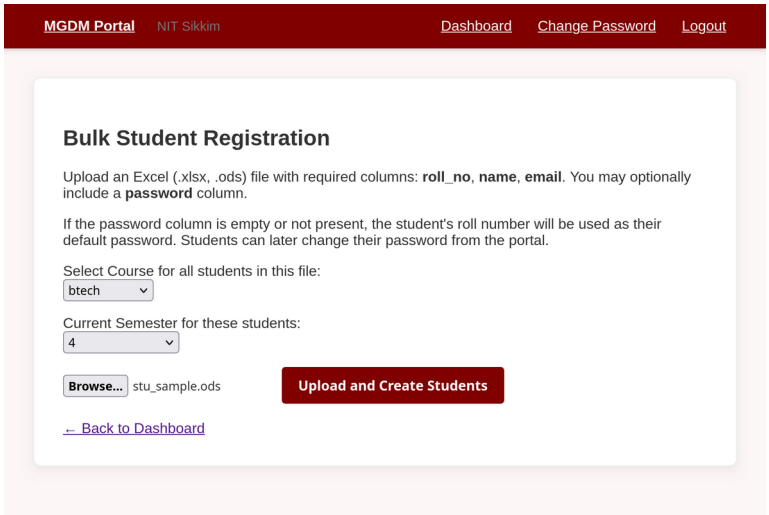
Admin Dashboard



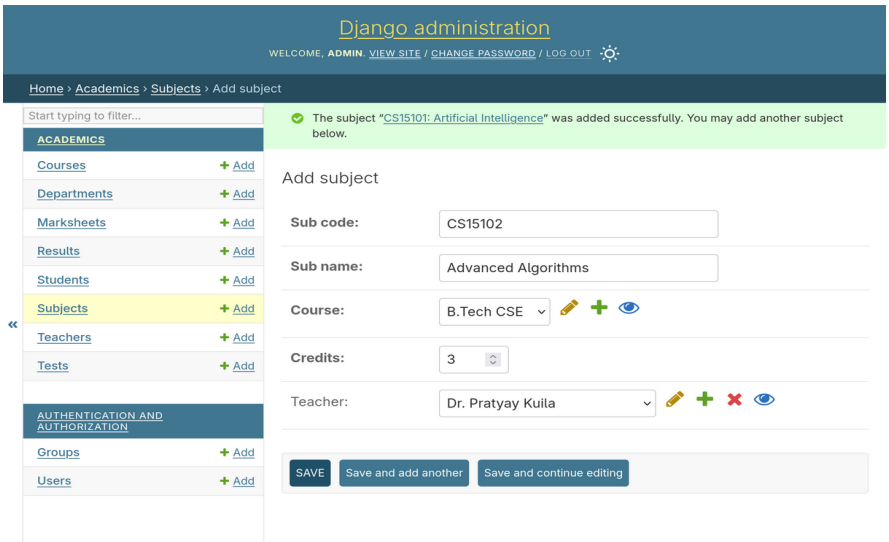
Marksheet Publishing by Admin



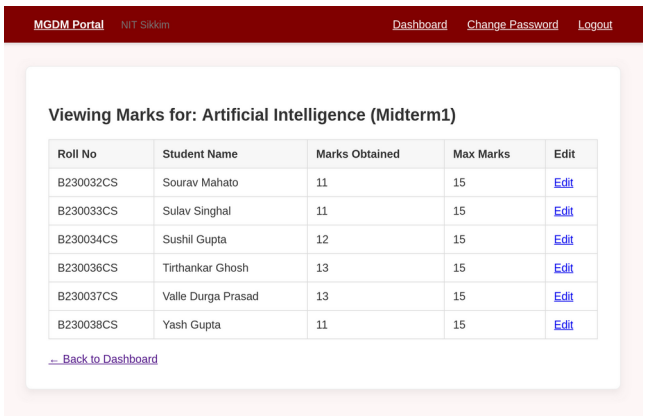
Option for Admin to Register Students in Bulk

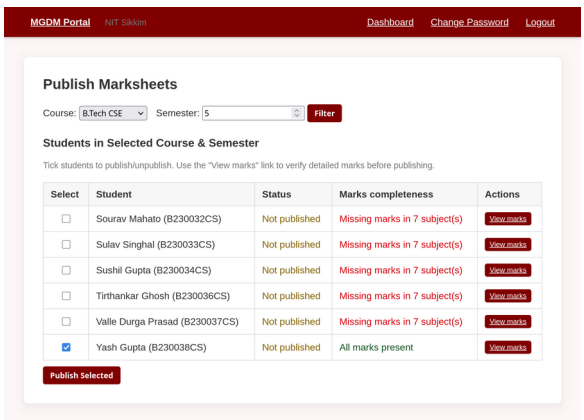


Django's administration panel

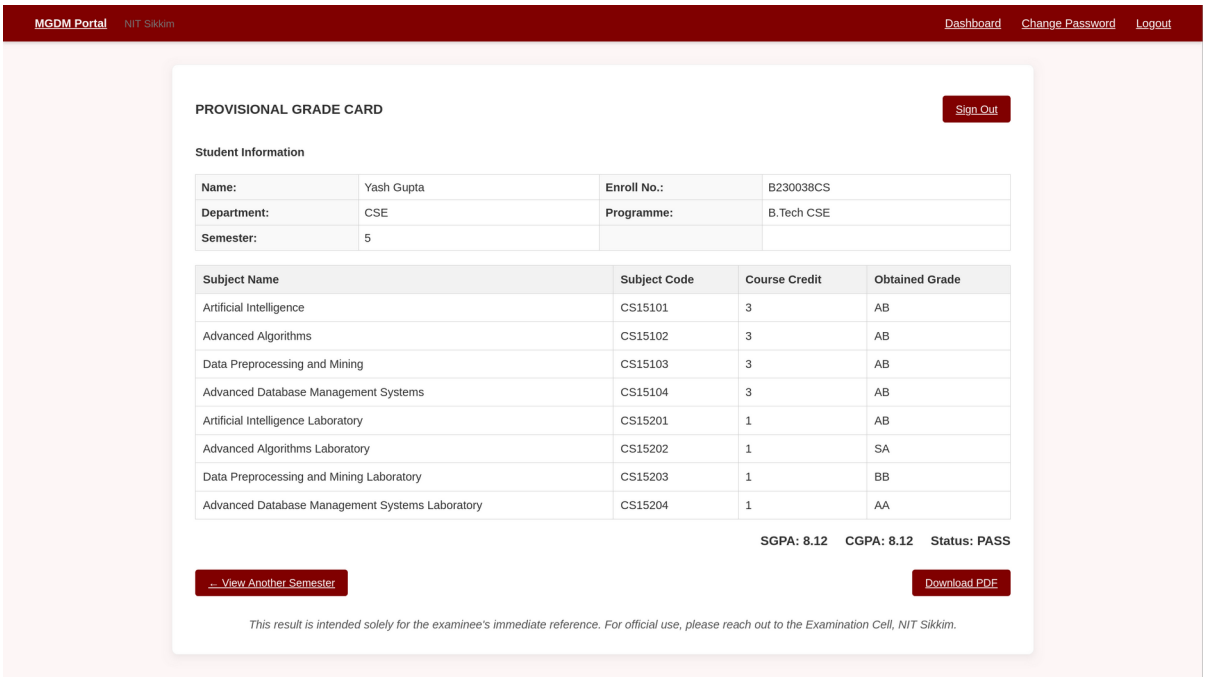


Teacher's view of publishing marks for tests:





Student’s view of viewing and downloading the Marksheet and it’s generated PDF



MGDM Portal NIT Sikkim Dashboard Change Password Logout

PROVISIONAL GRADE CARD

Sign Out

Student Information

Name:	Yash Gupta	Enroll No.:	B230038CS
Department:	CSE	Programme:	B.Tech CSE
Semester:	5		

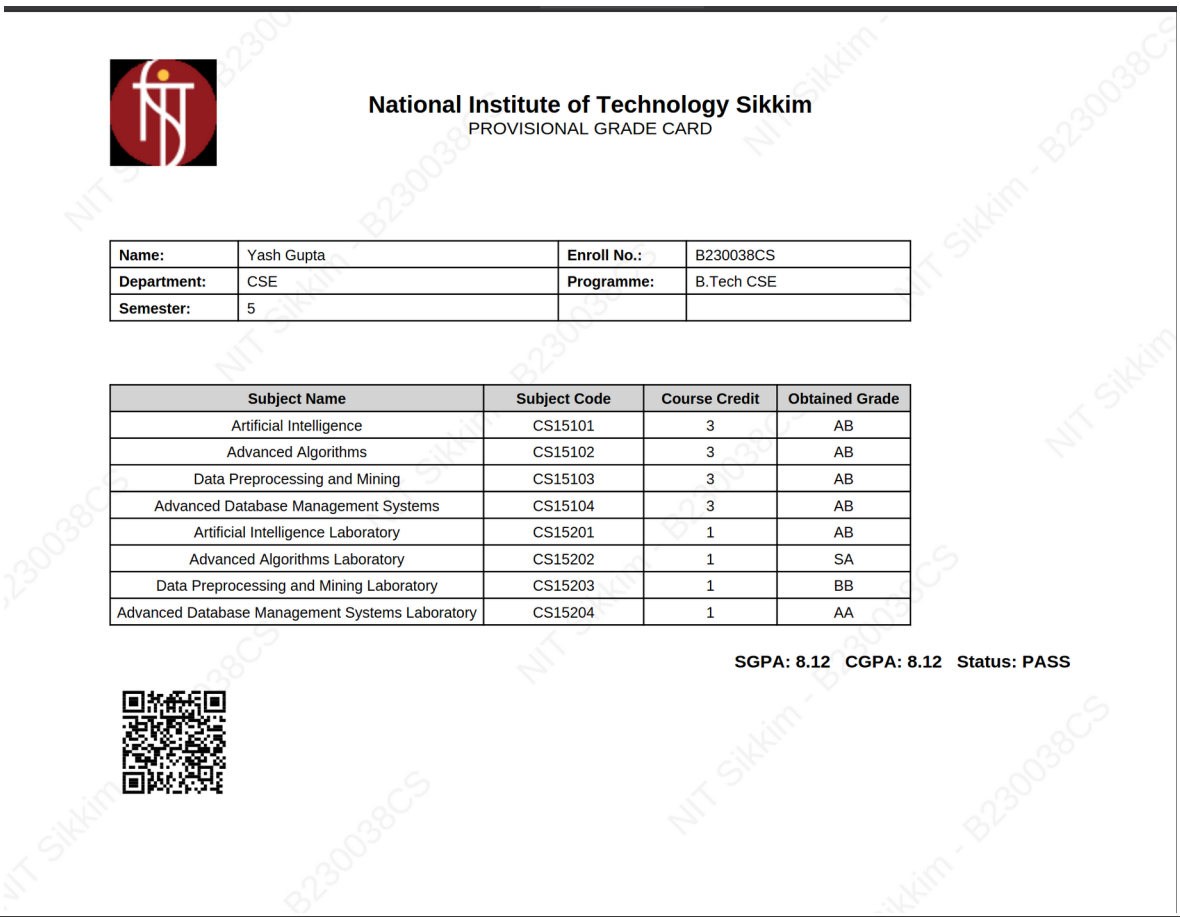
Subject Name	Subject Code	Course Credit	Obtained Grade
Artificial Intelligence	CS15101	3	AB
Advanced Algorithms	CS15102	3	AB
Data Preprocessing and Mining	CS15103	3	AB
Advanced Database Management Systems	CS15104	3	AB
Artificial Intelligence Laboratory	CS15201	1	AB
Advanced Algorithms Laboratory	CS15202	1	SA
Data Preprocessing and Mining Laboratory	CS15203	1	BB
Advanced Database Management Systems Laboratory	CS15204	1	AA

SGPA: 8.12 CGPA: 8.12 Status: PASS

← View Another Semester

Download PDF

This result is intended solely for the examinee's immediate reference. For official use, please reach out to the Examination Cell, NIT Sikkim.



National Institute of Technology Sikkim

PROVISIONAL GRADE CARD

Name:	Yash Gupta	**Enroll No.:**	B230038CS
Department:	CSE	**Programme:**	B.Tech CSE
Semester:	5		

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Artificial Intelligence	CS15101	3	AB
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Data Preprocessing and Mining	CS15103	3	AB
Advanced Database Management Systems	CS15104	3	AB
Artificial Intelligence Laboratory	CS15201	1	AB
Advanced Algorithms Laboratory	CS15202	1	SA
Data Preprocessing and Mining Laboratory	CS15203	1	BB
Advanced Database Management Systems Laboratory	CS15204	1	AA

SGPA: 8.12 CGPA: 8.12 Status: PASS

NIT Sikkim - B230038CS

6. CONCLUSION AND FUTURE SCOPE

CONCLUSION:

The “Online Student Marksheet Generation and Distribution Management System” has successfully developed and implemented, meeting all the core objectivves outlined previously. The project delivers a robust, secure and efficient platform automating entire process of marksheet generation and distribution, significantly reducing the manual effort required while putting a check to human errors that could crawl up in manual systems.

The clear separation of roles ensures data integrity, while features like bulk data upload and a formal publishing workflow makes the system a useful tool for integrating into existing academic workflow with minimal learning curve for all it’s users.

FUTURE SCOPE:

While the current system is fully functional, there are several aspects it could improve upon in future:

- OCR: Just like bulk excel file upload, a facility to just scan any printed or handwritten document, thereby getting rid of necessity to type data down on a spreadsheet if it’s present only in physical format.
- Password Recovery: Right now users can only change their passwords if they already remember it. If they forget their password then they need to contact the administrator. It could benefit from having a feature that allows users to recover/change their password by sending a code to email.
- Integration with other systems: Building APIs to integrate with other institutional systems such as student registrations thereby reducing redundancy of uploading student lists separately every semester.

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